

Day 8: Data Modeling Fundamentals

[DEF] Definition to Remember

Data Modeling in Power BI is the process of connecting multiple tables through relationship keys to enable accurate cross-table analytical querying and DAX measure evaluations.

1. Keys & Relationships

- * Primary Key (PK): A column containing unique values that uniquely identifies each entity in a dimension table (e.g., CustomerID).

2. Cardinality Types

1. One-to-Many (1:*) [Standard & Best Practice]: One unique record in the Dimension table matches multiple records in the Fact table.

3. Cross-Filter Direction

- * Single (One-Way): Filter flows from the 1-side (Dimension) down to the *-side (Fact). This is the optimal, performant default.

```
[Dim_Customer] (1)
      |
      | (Filter flows DOWN)
      |
[Fact_Sales] (*)
```

[MUST-WRITE] Must-Master Skills & Interview Points

1. Always prefer Single cross-filter direction over Bi-directional to avoid ambiguous relationship paths.
2. Know how to check relationship validity in Model View.
3. Understand why a clean relational model is 10x more important than writing complex DAX workarounds.

[RECALL] Quick Recall

Dimension (1) -> Fact (*) | Single Cross-Filter Direction | Clean PK-FK Relational Architecture